

Daily Combinatorics Drill #5

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Stars & Bars · Substitution Shifts · Inclusion–Exclusion · Counting with Caps.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Distributions and overlaps on sight. One formula each.

- How many pairs of non-negative integers (x, y) satisfy $x + y = 10$?
- In how many ways can 5 identical sweets be shared among 3 children (a child may get none)?
- Given $|A| = 15$, $|B| = 12$ and $|A \cap B| = 5$, find $|A \cup B|$.
- How many integers from 1 to 100 are divisible by 3?
- Two sets have $|A| = 8$, $|B| = 6$ and $|A \cup B| = 11$. Find $|A \cap B|$.
- How many triples of non-negative integers (x, y, z) satisfy $x + y + z = 4$?
- How many triples of *positive* integers (x, y, z) satisfy $x + y + z = 6$?
- In a class of 30, twenty study French, fifteen study German and eight study both. How many study neither?
- True or false: the number of ways to write 7 as an ordered sum of two positive integers is 6.
- In how many ways can 4 identical marbles be put into 2 boxes?

Section B Manipulation Drills

(≤50 s each)

Stars and bars, shifts, and two-set overlaps. Write the substitution down.

- How many triples of non-negative integers satisfy $x + y + z = 12$?
- How many triples of positive integers satisfy $x + y + z = 12$?
- Ten identical sweets are shared among 4 children so that each child gets at least one. In how many ways?
- How many integers from 1 to 1000 are divisible by neither 2 nor 5?

5. How many triples of non-negative integers satisfy $x + y + z = 10$ with $x \geq 2$?
6. In how many arrangements of the digits 1, 2, 3, 4 is the digit 1 *not* in the first position?
7. How many integers from 1 to 100 are divisible by 2 or by 3?
8. In how many ways can 8 be written as an ordered sum of three positive integers?
9. How many pairs of non-negative integers (x, y) satisfy $x + y \leq 5$?
10. How many 5-letter strings over the alphabet $\{A, B\}$ use both letters at least once?

Section C Substitution & Structure(≤ 90 s each)

Shift for lower bounds, include-exclude for upper bounds. State which you are using.

1. How many quadruples of positive integers satisfy $x + y + z + w = 15$ with no variable exceeding 5?
2. How many triples of non-negative integers satisfy $x + y + z = 15$ with every variable at most 6?
3. Of 100 students, 50 like tea, 40 like coffee, 30 like juice; 20 like tea and coffee, 15 tea and juice, 10 coffee and juice; 5 like all three. How many like none of the drinks?
4. How many integers from 1 to 1000 are divisible by 2, 3, or 5?
5. Eight identical apples are shared among 3 children so that no child gets more than 4. In how many ways?
6. How many arrangements of ABCDE have A not in the first position and B not in the second?
7. Three standard dice are rolled in order. How many outcomes have the three faces summing to 10?
8. How many positive integers less than 1000 have digit sum equal to 12?

Section D Challenge

(SMC / BMO1 difficulty)

Proof expected. Write the argument, not just the number.

1. Four letters are placed into four addressed envelopes, one per envelope. Show, using inclusion-exclusion, that in exactly 9 placements no letter is in its correct envelope.
2. Prove that the number of ways to write n as an *ordered* sum of positive integers is 2^{n-1} .
3. How many integers from 1 to 360 share no common factor with 360 other than 1? Justify via inclusion-exclusion.

4. Five students each join exactly one of three clubs, and every club must gain at least one member. Show that there are exactly 150 ways this can happen.
5. Prove that exactly 998 910 of the integers from 1 to 10^6 are neither perfect squares nor perfect cubes.

“Overcount boldly, then subtract with precision — inclusion–exclusion is bookkeeping for the brave.”

other prep to look at today:
Daily Integral for Calculus and Logic